

Safi Ullah

AI & Machine Learning Engineer | Computer Vision Practitioner

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PROFESSIONAL SUMMARY

Computer Science student (CGPA 3.94/4.0) at the University of Layyah with practical experience building end-to-end machine learning pipelines, CNN-based image classifiers, and real-time computer vision applications. Proficient in TensorFlow, Keras, Scikit-Learn, OpenCV, MediaPipe, and Flask REST API deployment. 7+ published AI/ML projects on GitHub demonstrating engineering capability across deep learning, predictive modeling, and computer vision domains.

EDUCATION

BSc Computer Science — In Progress

University of Layyah, Hafizabad | 2023 – Present | CGPA: 3.94/4.0 (3 semesters)

Intermediate in Computer Science (ICS)

D.G Khan Board, Punjab | 2021 – 2023 | Score: 990/1100

Matriculation (Science)

D.G Khan Board, Punjab | 2019 – 2021 | Score: 1067/1100

TECHNICAL SKILLS

Programming: Python 3, SQL, HTML/CSS/JavaScript

Machine Learning: Scikit-Learn, Logistic Regression, Random Forest, Gradient Boosting, Feature Engineering

Deep Learning: TensorFlow, Keras, CNN, Transfer Learning (MobileNetV2), Grad-CAM, Data Augmentation

Computer Vision: OpenCV, MediaPipe (Hand Tracking, Face Mesh), ALPR, Real-time Processing

Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, Imbalanced Data Handling

Web & Deployment: Flask, FastAPI, Streamlit, REST APIs, Docker, Gunicorn

Tools: Git, GitHub, VS Code, Jupyter Notebook, Linux CLI

KEY PROJECTS

Brain Tumor MRI Classifier (NeuroScan AI)

TensorFlow · Keras · MobileNetV2 · Grad-CAM · FastAPI · Python

Research-grade CNN classifying brain MRI scans as Tumor/No-Tumor. Transfer learning with MobileNetV2. 95.2% test accuracy, 95.7% recall, 0.98 ROC-AUC. Grad-CAM visual explainability heatmaps. FastAPI REST backend with web UI. Educational/research purpose only — not a clinical tool. github.com/SAFIULLAH012/Tumor_VS-NON_tumor

Framingham CHD Risk Prediction

Scikit-Learn · Logistic Regression · Pandas · Streamlit

10-year CHD risk binary classifier on 4,238 patients. Three models compared with 5-fold stratified CV. Logistic Regression selected: highest recall (0.58) and ROC-AUC (0.70). Correct metric reporting for imbalanced data. github.com/SAFIULLAH012/heart-disease-prediction

OpenCV & MediaPipe Vision Lab

OpenCV · MediaPipe · Python · Tesseract

Real-time computer vision suite: ALPR license plate detection, 3D hand tracking (21 landmarks at 25+ FPS), face mesh (468 landmarks). github.com/SAFIULLAH012/opencv_learning

Price Prediction Models (Flight / House / Laptop)

Scikit-Learn · Random Forest · Pandas · Feature Engineering

Three separate end-to-end regression pipelines. R^2 scores: Flight ~0.81, House ~0.78, Laptop ~0.85. Strong feature engineering pipelines for each domain.

Cat vs Dog CNN Classifier

TensorFlow · Keras · CNN

Binary CNN image classifier with dropout and data augmentation. ~90% test accuracy.
github.com/SAFIULLAH012/CAT_DOG_CLASSIFIER

GITHUB

All projects: github.com/SAFIULLAH012 | LinkedIn: linkedin.com/in/safiullah012